Worksheet 4B: What Derivatives and Integrals Tell Us

What This Worksheet Is About

We've learned *how* to compute derivatives and integrals. Now: *what do they mean*?

- **Derivatives** → rate of change, slope, velocity, increasing/decreasing
- **Integrals** → accumulation, area, total distance, total quantity

These interpretations are why calculus is so powerful!

Part A: Derivatives — Rate of Change and Slope (16 problems)

Derivative	Meaning
$f'(x) > 0$	Function increasing
$f'(x) < 0$	Function decreasing
$f'(x) = 0$	Horizontal tangent (potential max/min)
$f''(x) > 0$	Concave up
$f''(x) < 0$	Concave down

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1. For $f(x) = x^2$, $f'(x) = 2x$.

- (a) Where is f increasing? Decreasing?
- (b) Where is the tangent horizontal?
- (c) What is the slope at $x = 3$?

2. For $g(x) = x^3 - 3x$:

- (a) Find $g'(x)$.
- (b) Find critical points (where $g'(x) = 0$).
- (c) Classify as local max or min.

3. For $h(t) = t^4 - 8t^2$:

- (a) Find all critical points.
- (b) Use the first or second derivative test to classify each.

4. Find the tangent line to $f(x) = x^2$ at $(2, 4)$.

5. Find the tangent line to $g(x) = x^3 - x$ at $x = 1$.

6. At what point(s) on $y = x^3 - 12x$ is the tangent horizontal?

Physics Applications:

7. Height: $h(t) = 20t - 5t^2$ meters.

- (a) Find velocity $v(t) = h'(t)$.
- (b) When is velocity zero? What's happening then?
- (c) Find acceleration $a(t) = h''(t)$. Interpret it.

8. Position: $s(t) = t^3 - 6t^2 + 9t$ meters.

- (a) Find velocity and acceleration.
- (b) When is the object at rest?
- (c) When is acceleration zero?

Second Derivative and Concavity:

9. For $f(x) = x^3$: - Where is $f''(x) > 0$? (Concave up) - Where is $f''(x) < 0$? (Concave down) - Where is the inflection point?

10. For $g(x) = x^4 - 6x^2$: - Find inflection points. - Describe concavity on each interval.

Optimization:

11. Profit: $P(x) = -x^2 + 80x - 1200$. - Find the quantity maximizing profit. - Find maximum profit.

12. Area with fixed perimeter: A rectangle has perimeter 100m. - Express area as $A(w) = w(50 - w) = 50w - w^2$. - Find dimensions maximizing area.

13. A ball's height is $h(t) = -5t^2 + 30t + 10$. - When does it reach maximum height? - What is the maximum height?

14. Revenue: $R(p) = 500p - 2p^2$. - Find price maximizing revenue. - Find maximum revenue.

15. Cost: $C(q) = q^3 - 12q^2 + 60q + 100$. - Find marginal cost $C'(q)$. - At what quantity is marginal cost minimized?

16. Summary: The derivative being zero tells us _____, and the second derivative's sign tells us _____.

Part B: Integrals — Accumulation and Area (16 problems)

Integral	Meaning
$\int_a^b f(x)\,dx$	Signed area under curve
$\int v(t)\,dt$	Displacement
$\int r(t)\,dt$	Total quantity accumulated

Area Under Curves:

17. Find the area under $f(x) = 3$ from $x = 1$ to $x = 5$. - Geometrically: rectangle with base _____ and height _____ - By integral: $\int_1^5 3\,dx =$

18. Find the area under $f(x) = x$ from $x = 0$ to $x = 4$. - Geometrically: triangle - By integral: $\int_0^4 x\,dx =$

19. $\int_0^2 x^2\,dx =$

20. $\int_1^3 (2x+1)\,dx =$

Signed Area:

21. $\int_{-2}^2 x\,dx =$ _____

Why? (Think about symmetry)

22. $\int_{-1}^1 x^3\,dx =$ _____

This is an odd function on a symmetric interval.

23. $\int_{-2}^2 x^2\,dx =$ _____

This is an even function: $\int_{-a}^a f(x)dx = 2\int_0^a f(x)dx$

24. For $f(x) = x - 1$ from $x = 0$ to $x = 2$: - Sketch the function. - Where is it positive? Negative? - Calculate $\int_0^2 (x-1)dx$. Interpret the result.

Physical Applications:

25. Velocity: $v(t) = 3t^2$ m/s. Find displacement from $t = 0$ to $t = 2$.

26. Velocity: $v(t) = t-2$ m/s from $t = 0$ to $t = 4$. - (a) When is the object moving forward? Backward? - (b) Find net displacement: $\int_0^4 (t-2)dt$ - (c) Find total distance traveled.

27. Flow rate: $r(t) = 100 - 4t$ liters/min. - Find total flow from $t = 0$ to $t = 10$.

28. Population growth rate: $P'(t) = 200 + 10t$ people/year. - Find total growth over 5 years.

Accumulation Functions:

29. Let $A(x) = \int_0^x t^2\,dt$. - (a) Compute $A(x)$ explicitly. - (b) Find $A(1)$, $A(2)$, $A(3)$. - (c) Find $A'(x)$. What do you notice?

30. If $F(x) = \int_1^x (t^3+1)dt$, what is $F'(x)$?

(Use the Fundamental Theorem — no need to compute the integral!)

31. If $G(x) = \int_0^x \sqrt{t}\,dt$, what is $G'(x)$?

32. Explain: "Integrating then differentiating gets you back to where you started."

Part C: The Fundamental Theorem — Tying It Together (12 problems)

The Fundamental Theorem of Calculus:

1. $\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x)$
2. $\int_a^b f(x)\,dx = F(b) - F(a)$ where $F' = f$

Using Part 1:

33. $\frac{d}{dx}\left[\int_0^x t^4\,dt\right] =$

34. $\frac{d}{dx}\left[\int_2^x (3s^2 - s)\,ds\right] =$

35. $\frac{d}{dx}\left[\int_1^x \frac{1}{t^2}\,dt\right] =$

Using Part 2:

36. $\int_0^3 2x\,dx =$ (use antiderivative x^2)

37. $\int_1^4 (x^2 - 1)\,dx =$

38. $\int_0^8 x^{1/3}\,dx =$

39. $\int_1^e \frac{1}{x}\,dx =$ (antiderivative is $\ln x$; we'll derive this in 4D!)

Round-Trip Verification:

40. Start with $f(x) = x^3$. - Integrate: $\int f(x)dx =$ - Differentiate your answer. - Did you get $f(x)$ back?

41. Start with $g(x) = 6x^2 - 4x$. - Differentiate: $g'(x) =$ - Integrate your answer. - Did you get $g(x)$ back (plus C)?

42. Explain the asymmetry: - Differentiating then integrating gives $f(x) + C$ - Integrating then differentiating gives exactly $f(x)$

Why the difference?

Connecting Position, Velocity, Acceleration:

43. Complete the chain:

Position $s(t)$ $\xrightarrow{\text{derivative}}$ _____ $\xrightarrow{\text{derivative}}$ _____

_____ $\xrightarrow{\text{integral}}$ Velocity $\xrightarrow{\text{integral}}$ _____

44. An object has acceleration $a(t) = 6t$, with $v(0) = 2$ and $s(0) = 1$. - Find $v(t)$ by integrating $a(t)$ and using the initial condition. - Find $s(t)$ by integrating $v(t)$ and using the initial condition.

Part D: Area Between Curves (8 problems)

Area between $f(x)$ and $g(x)$ from a to b:

$$\text{Area} = \int_a^b |f(x) - g(x)|\, dx$$

If $f(x) \geq g(x)$ on $[a, b]$: Area $= \int_a^b (f(x) - g(x))\, dx$

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45. Find area between $y = x^2$ and $y = x$ from $x = 0$ to $x = 1$. - Which is on top? - Set up and compute the integral.

46. Find area between $y = x^2$ and $y = 4$. - Find intersection points. - Set up and compute.

47. Find area enclosed by $y = x^2$ and $y = 2x$. - Find intersections. - Integrate from intersection to intersection.

48. Find area between $y = x^3$ and $y = x$ from $x = -1$ to $x = 1$. - Careful: which is on top changes at $x = 0$!

Applications:

49. Consumer Surplus: Demand is $p = 50 - 2q$, equilibrium price is $p^* = 30$. - Consumer surplus $= \int_0^{q^*} (50 - 2q - 30)dq$ - Find q^* and calculate surplus.

50. The area between a velocity curve and the t-axis represents _____.

51. The area between two position curves represents the _____ between them.

52. Gini Coefficient: Perfect equality is $y = x$. A Lorenz curve is $y = x^2$. - Gini $= \frac{\text{Area between curves}}{\text{Area under } y=x}$ - Calculate it.

Part E: Connecting to ML and Optimization (8 problems)

53. Loss function: $L(w) = w^2 - 6w + 10$. - Find $L'(w)$. - Find the minimum. - Verify with second derivative.

54. Gradient descent update: $w_{new} = w_{old} - \alpha \cdot L'(w_{old})$

For $L(w) = w^2 - 4w + 5$ with $\alpha = 0.1$, starting at $w_0 = 0$: - Calculate w_1, w_2, w_3. - What value is w approaching?

55. Total loss over dataset $[0, 4]$ with loss function $\ell(x) = (x - 2)^2$:

Total $= \int_0^4 (x - 2)^2 dx$. Calculate it.

56. Mean loss $= \frac{1}{4} \int_0^4 (x - 2)^2 dx$. Calculate it.

57. Regularized loss: $L(w) = (w-5)^2 + 0.5w^2$. - Expand and simplify. - Find optimal w. - Compare to unregularized optimum $w = 5$.

58. Explain: Why is finding where $L'(w) = 0$ central to training ML models?

59. Explain: Why might we care about $\int L(w)dw$ over a region of weight space?

60. The Fundamental Theorem connects local information (derivative at a point) to global information (integral over an interval). How does this relate to: - Local vs. global minima in optimization? - Gradient (local) vs. total loss (global)?